

SCRAP SPACE

Roman Rabbat, Graham Speight, Garrett Comes, Joseph Moskovitz, Kevin Gao, Samuel Gatewood, Swaroop Sathyanarayanan, Anish Kulkarni, Jacob J Menssen, Cuong Nguyen

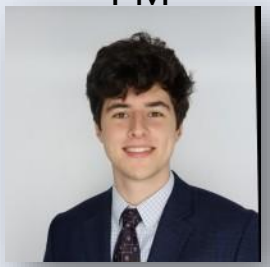
The Team



Roman Rabbat
Primary Investigator



Kevin Gao
Bus Components PM



Samuel Gatewood
De-orbit PM



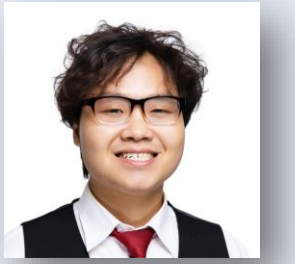
Joseph Moskovitz
Economic Model PM



Jacob Menssen
Operations PM



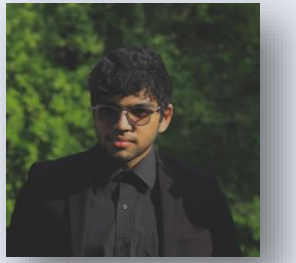
Graham Speight
Prop & Bus system Engineer



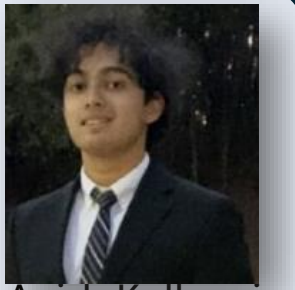
Cuong Nguyen
De-Tumble Engineer



Garrett Comes
GNC & De-Tumble PM



Swaroop Sathyanarayanan
De-orbit Engineer



Anish Kulkarni
De-orbit Engineer



Dr Carr
Advisor

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Problem

Space clutter is becoming a larger issue every year, and cascading collision is a growing risk in a thriving LEO market.

Mission Statement

Develop an autonomous space system to safely de-tumble and de-orbit space debris

Our objectives:

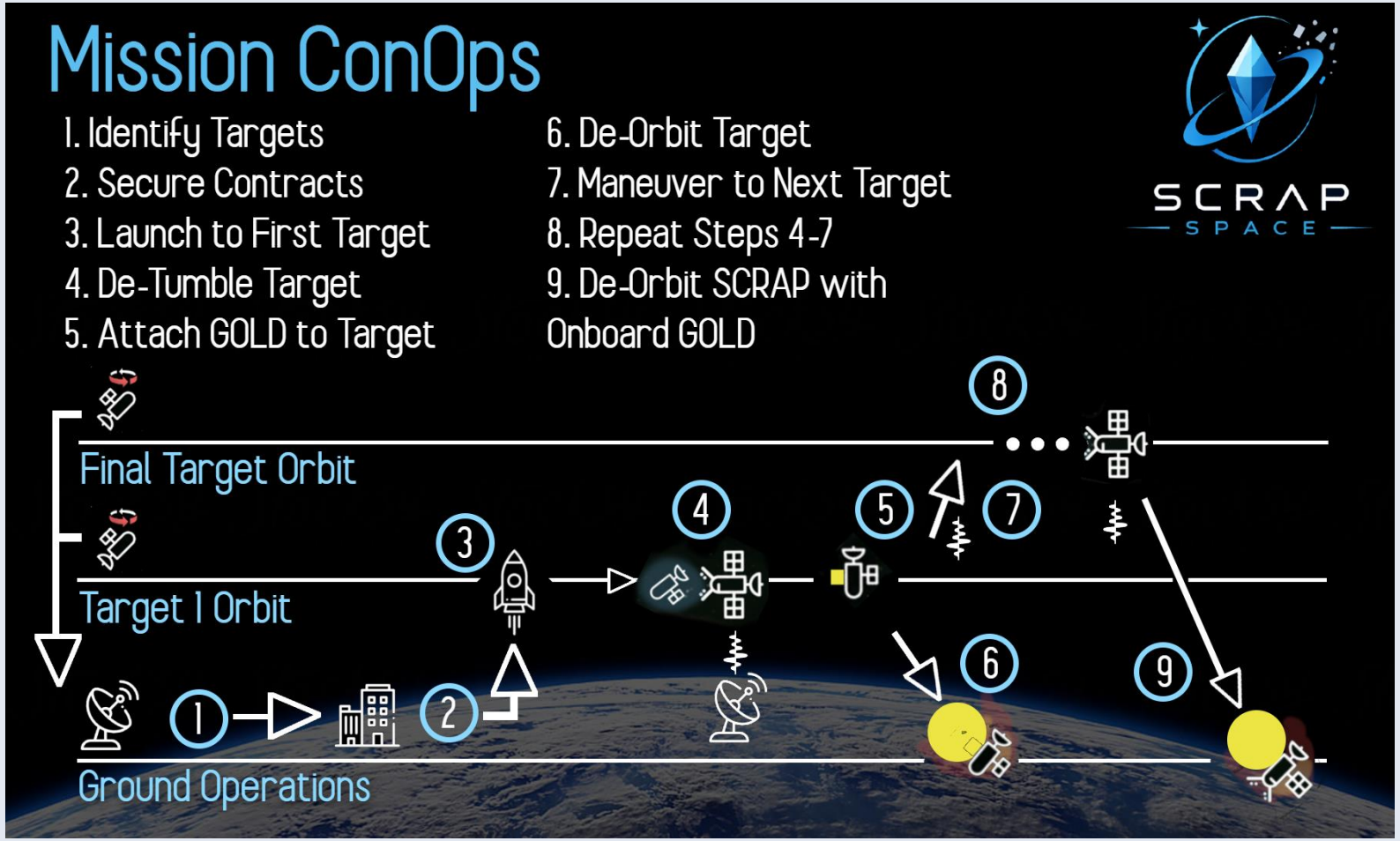
1. Design the hardware
   - Capable of de-tumbling and de-orbiting at least one >100 kg piece of debris within its operating range
2. Build the software
   - Capable of identifying and tracking candidates for de-orbit, provided position and velocity information for all debris in publicly available sources
3. Develop the protocol
   - Capable of identifying orbits within LEO that are ideal for targeting multiple objects

Mission Requirements

1. The system shall **identify**, **categorize**, and **score** candidate objects by mass, form-factor, relative orbit density, and rotational energy to determine viable de-orbit mission architectures
2. The system shall **de-orbit** target objects safely and
    1. The de-tumbling system shall de-tumble identified target objects from their initial rotational velocity to a final maximum total rotational velocity of 5 deg/sec within 2 weeks
3. The system shall operate **autonomously** throughout constellation de-orbit operations
4. The propulsion system shall maintain station-keeping in between constellations and provide attitude and pointing control during de-orbit operations



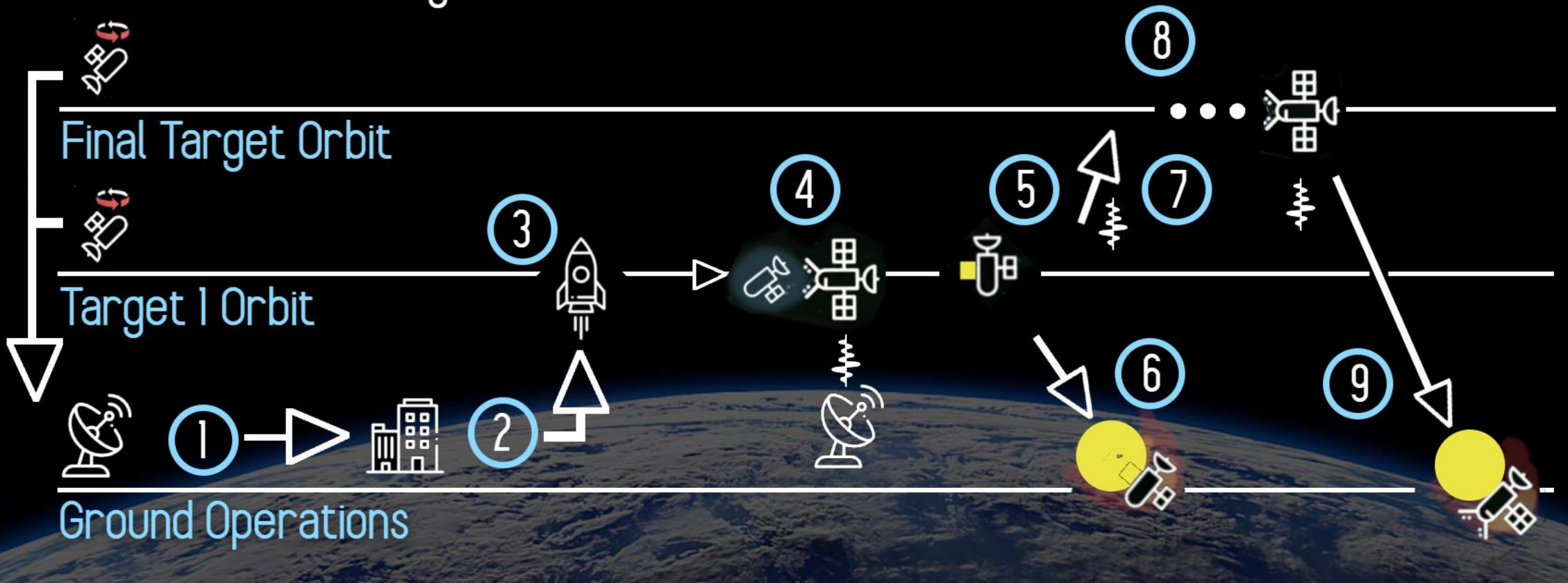
Con-Ops Diagram



Mission ConOps

1. Identify Targets
2. Secure Contracts
3. Launch to First Target
4. De-Tumble Target
5. Attach GOLD to Target
6. De-Orbit Target
7. Maneuver to Next Target
8. Repeat Steps 4-7
9. De-Orbit SCRAP with Onboard GOLD

SCRAP SPACE



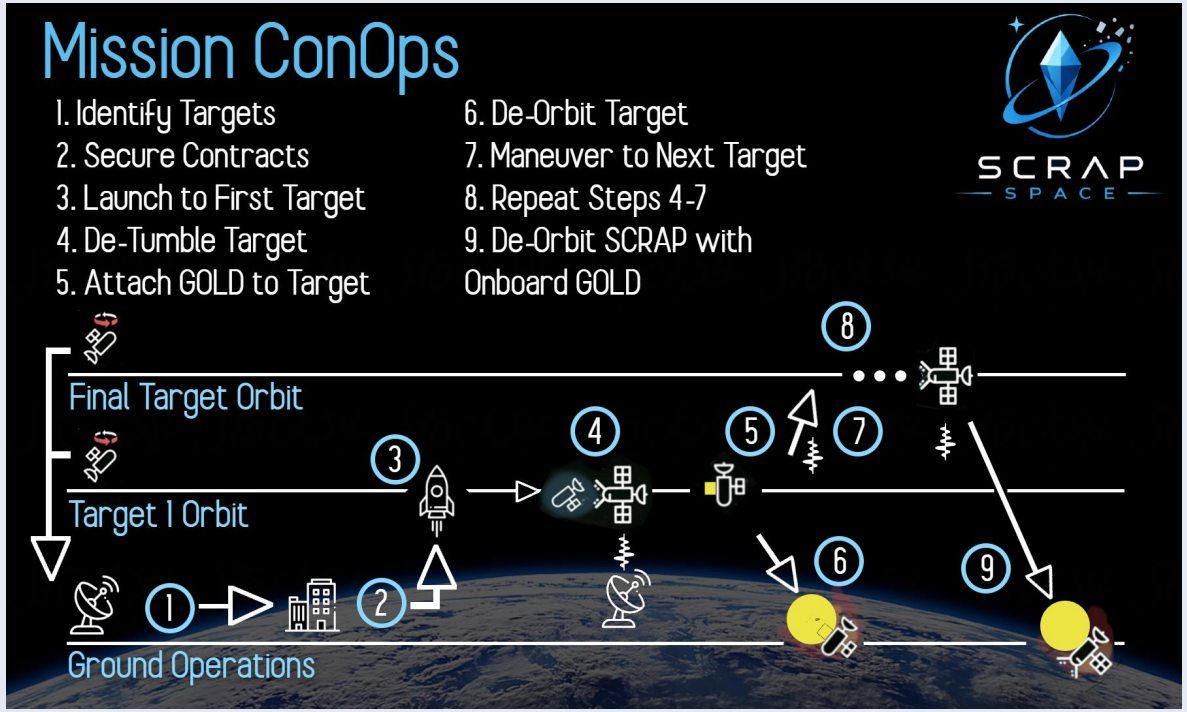
Operational Constraints

Altitude
- Upper limit of ~700km

Satellite Selection
- Extremely computationally demanding to find ideal paths and targets.
- Selection will be heavily determined by negotiations with parties of interest.

Magnet
- Quenching the magnet will completely disable it and cause a mission failure.
- Must limit current, temperature, and B field.

GOLD Attachment
- Relatively flat surface needed.
- Surface below 120 °C
- Normal contact.



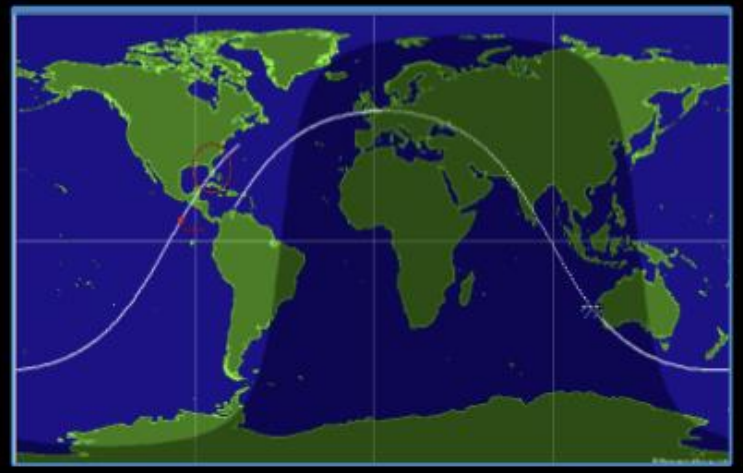
Mission Operations: Command Console



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SCRAP MISSION OPERATIONS 4/16/2026 - 14:34:32





Next Downlink:

T - 00:17:38

Mission Timeline



GOLD Attachment Checklist

- ❑ Visual Rotation Confirmation
- ❑ Surface Acceptance
- ❑ Surface Temperature Good
- ❑ Power Nominal
- ❑ Simulated 2σ Success Rate

Subsystem Status

ADCS: Nominal
COMM: Nominal
Magnet: Nominal
Processing: Nominal
Thermal: + 18C
Bus Power: 85%

IR Camera: Nominal
Camera: Nominal
Reaction Wheel A: 6258 RPM
Reaction Wheel B: 5539 RPM
Reaction Wheel C: 7848 RPM
Reaction Wheel D: 0 RPM



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Communications Protocol

Band

Ground Station

- Private ground station for normal mission operations
- AWS Ground Station support for critical mission stages

Packet Protocol

- CCSDS Standards (Blue Books)

Communications Packets

Uplink

- *Proceed*
  - After satisfaction of “Go” checklist
- *Wait*
  - Put SCRAP into power safe mode for further analysis
- *Maneuver*
  - Command orbital maneuver
- *Request*
  - Request Telemetry, Logs, or Video

Downlink

- *Health*
  - Bus voltage, solar array current, temperature, fault flags
- *Telemetry*
  - Health + detailed time history of data
  - Mission-relevant data
- *Logs*
  - Full mission-relevant data
- *Video*
  - Video of target



Mission Operations: Fault Management

Mission Operations: Go / No Go

Economic Justification

Market Opportunity

- LEO satellite market projected to surpass $24B by 2032 (Starlink & Project Kuiper)
- Operators are paying higher insurance premiums and facing stricter launch approval processes

Regulatory Impetus

- FCC now mandates 5-year de-orbit compliance, non-compliance means no future launch licenses
- ESA requires >90% disposal success rate for new missions

Business Model: Dual-Use Revenue

Civil / Intergovernmental (ESA, UN COPUOS)

- Non-dilutive foundational revenue, flight heritage, global legitimacy

US Department of Defense (Space Force)

- Contactless attitude control on uncooperative objects as a strategic defense capability
- Directly aligned with USSF dynamic space operations doctrine

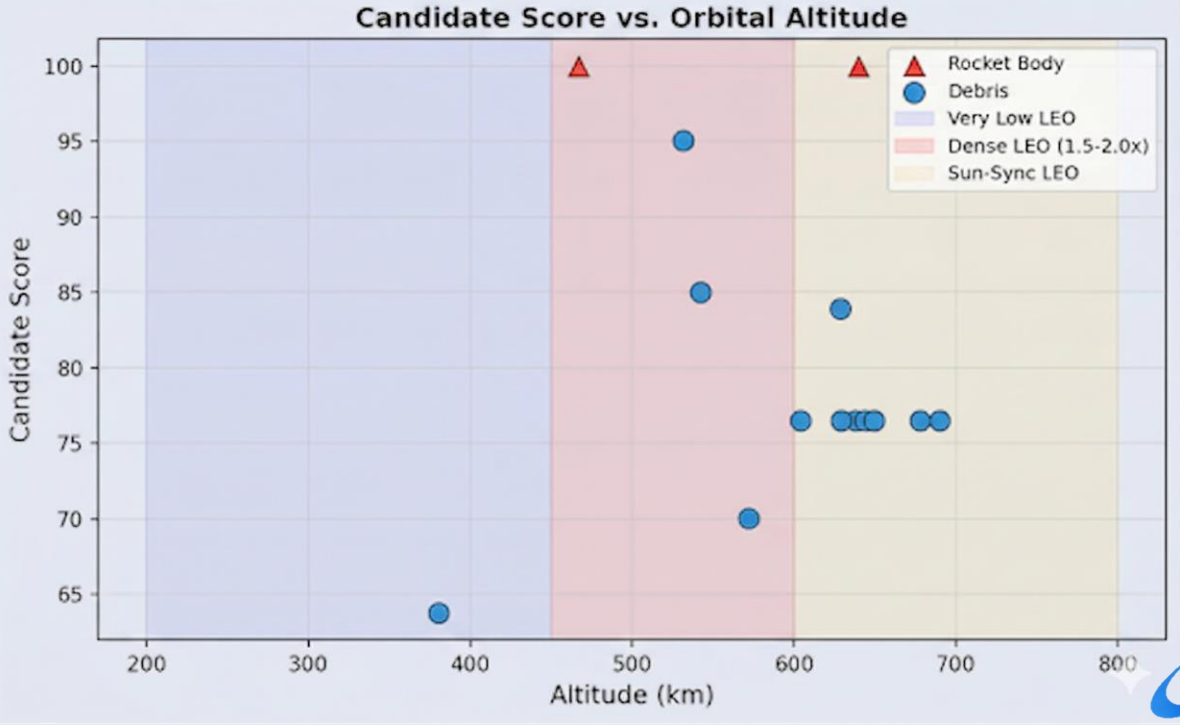
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Satellite Selection

- Custom pipeline cross-references Space-Track, DISCOS web, and GCAT catalogs ~**29,000 LEO objects**
- Altitude band multiplier prioritizes congested orbits
- Rocket bodies and long-lived debris score highest
- Numerical decay propagation with NRLMSISE-00 atmosphere model and J2/full gravity
- Top candidates: SL-3 rocket body, N-2 rocket body

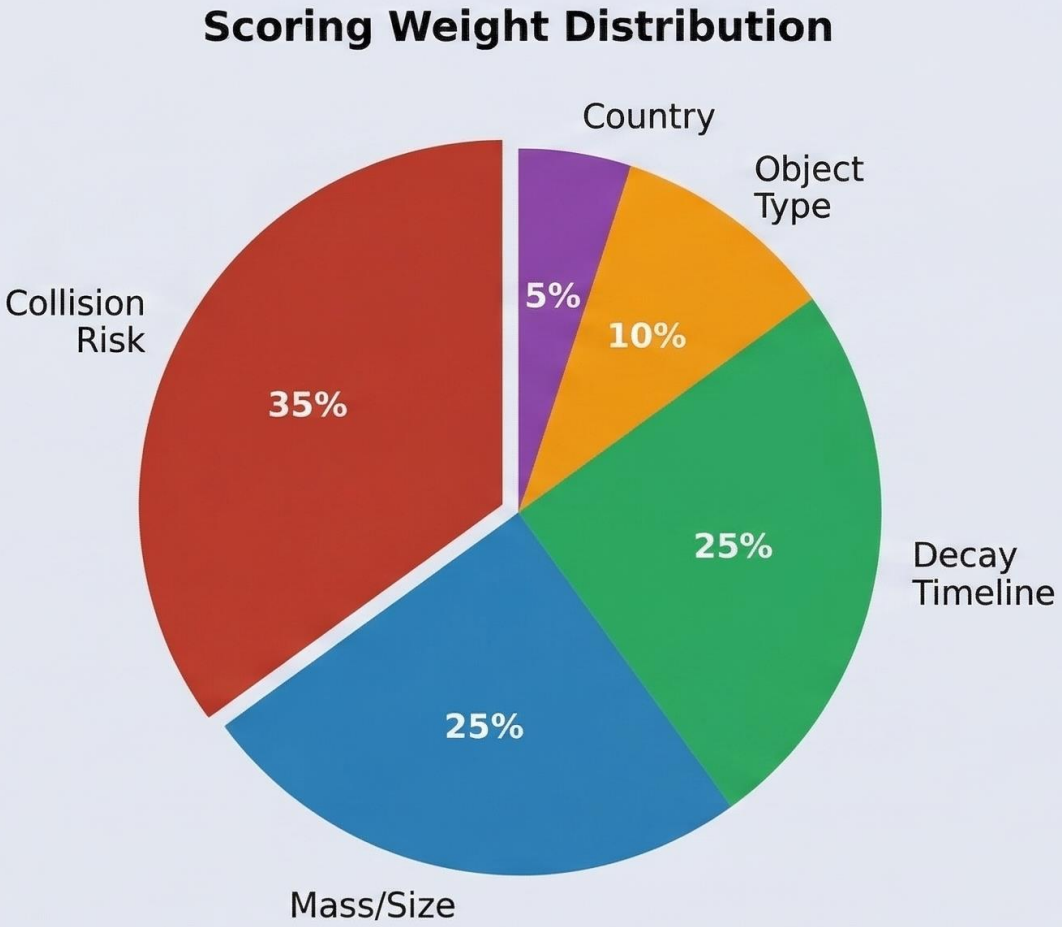
Debris Removal Candidate Ranking

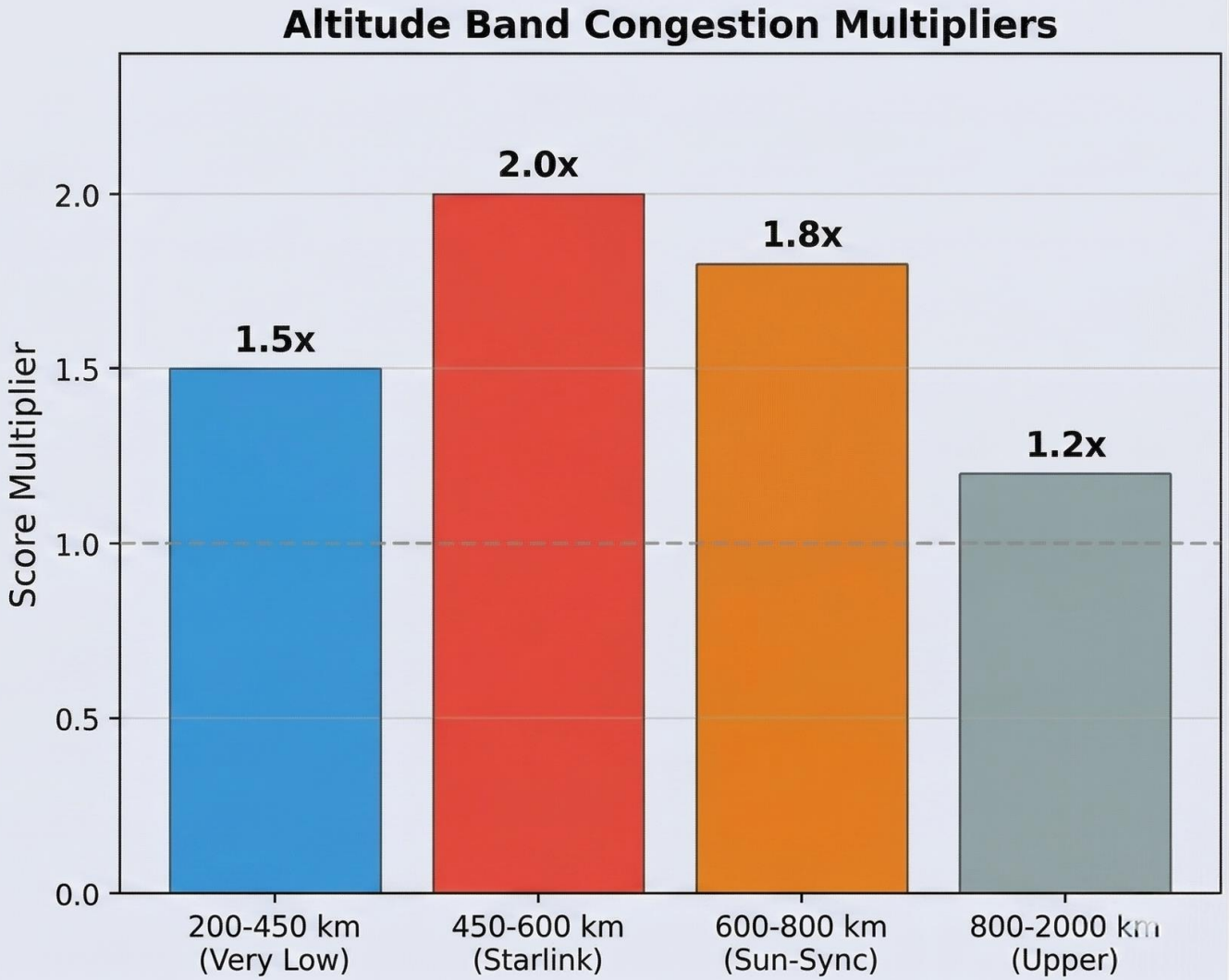
Rank	Name	Type	Alt (km)	Mass (kg)	Orbital Band	Score
1	SL-3 R/B	Rocket Body	467	1427	Dense LEO (Starlink)	100.0
2	N-2 R/B	Rocket Body	640	892	Sun-Sync LEO	100.0
3	FALCON 9 DEB	Debris	532	100	Dense LEO (Starlink)	95.1
4	COSMOS 2251 DEB	Debris	542	N/A	Dense LEO (Starlink)	85.0
5	SL-24 DEB	Debris	629	50	Sun-Sync LEO	83.9
6	CZ-6A DEB	Debris	638	N/A	Sun-Sync LEO	76.5
7	FENGYUN 1C DEB	Debris	644	N/A	Sun-Sync LEO	76.5
8	COSMOS 2251 DEB	Debris	629	N/A	Sun-Sync LEO	76.5
9	SL-8 DEB	Debris	650	N/A	Sun-Sync LEO	76.5
10	CZ-6A DEB	Debris	690	N/A	Sun-Sync LEO	76.5
11	FENGYUN 1C DEB	Debris	604	N/A	Sun-Sync LEO	76.5
12	SL-8 DEB	Debris	678	N/A	Sun-Sync LEO	76.5
13	CZ-2C DEB	Debris	572	1	Dense LEO (Starlink)	70.0
14	COSMOS 886 DEB	Debris	381	N/A	Very Low LEO	63.7



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Satellite Selection



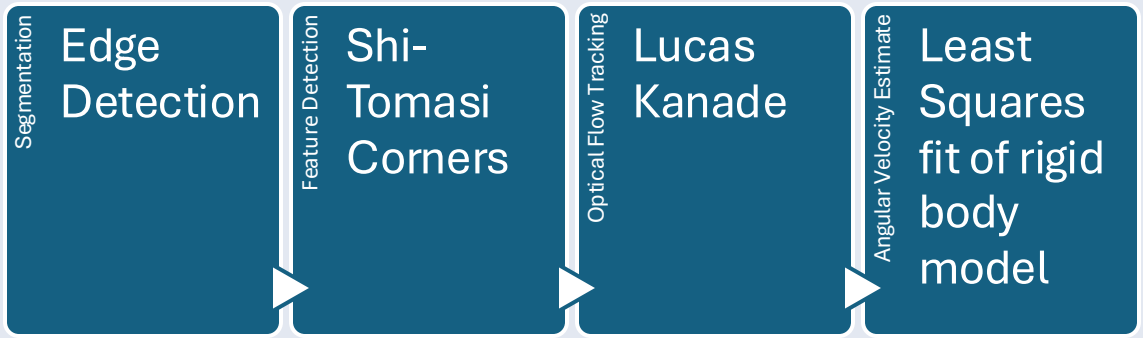


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GNC: Vision Tracking

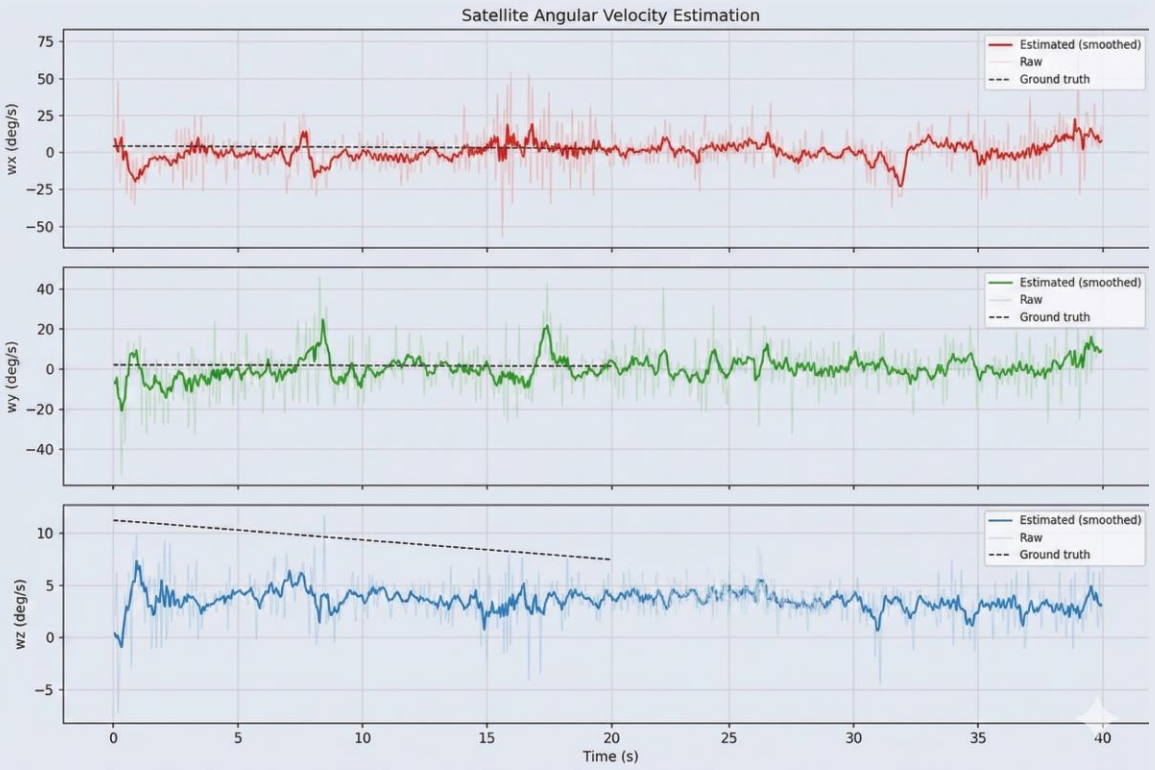
- Computer vision driven from an onboard FPGA will determine angular momentum vectors
- Pipeline approach allows consistent time between estimates with low power usage

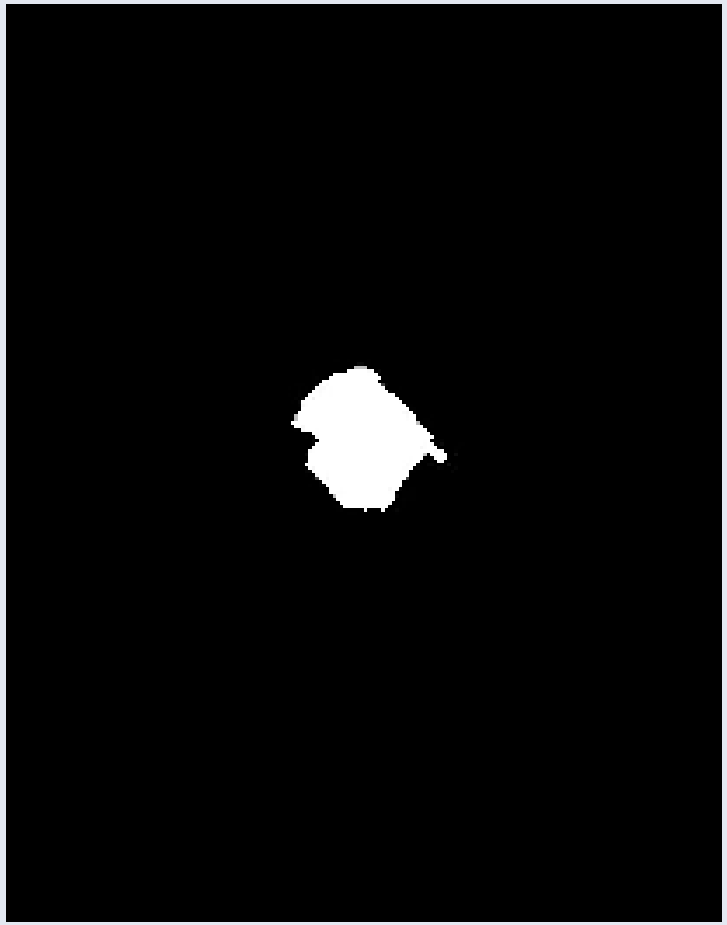


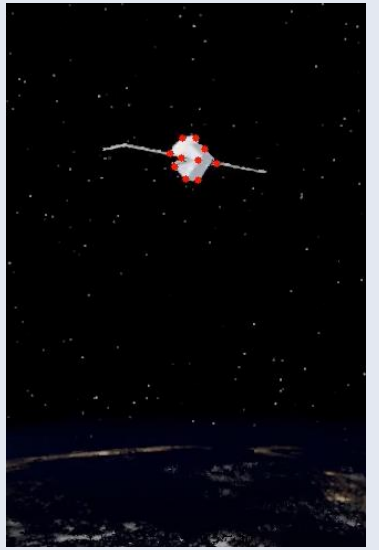


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GNC: Vision Tracking





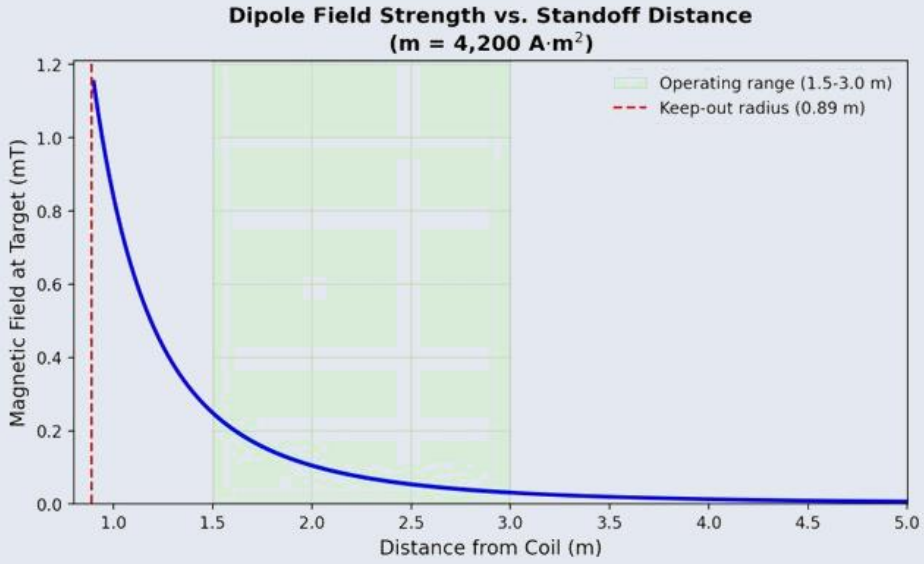


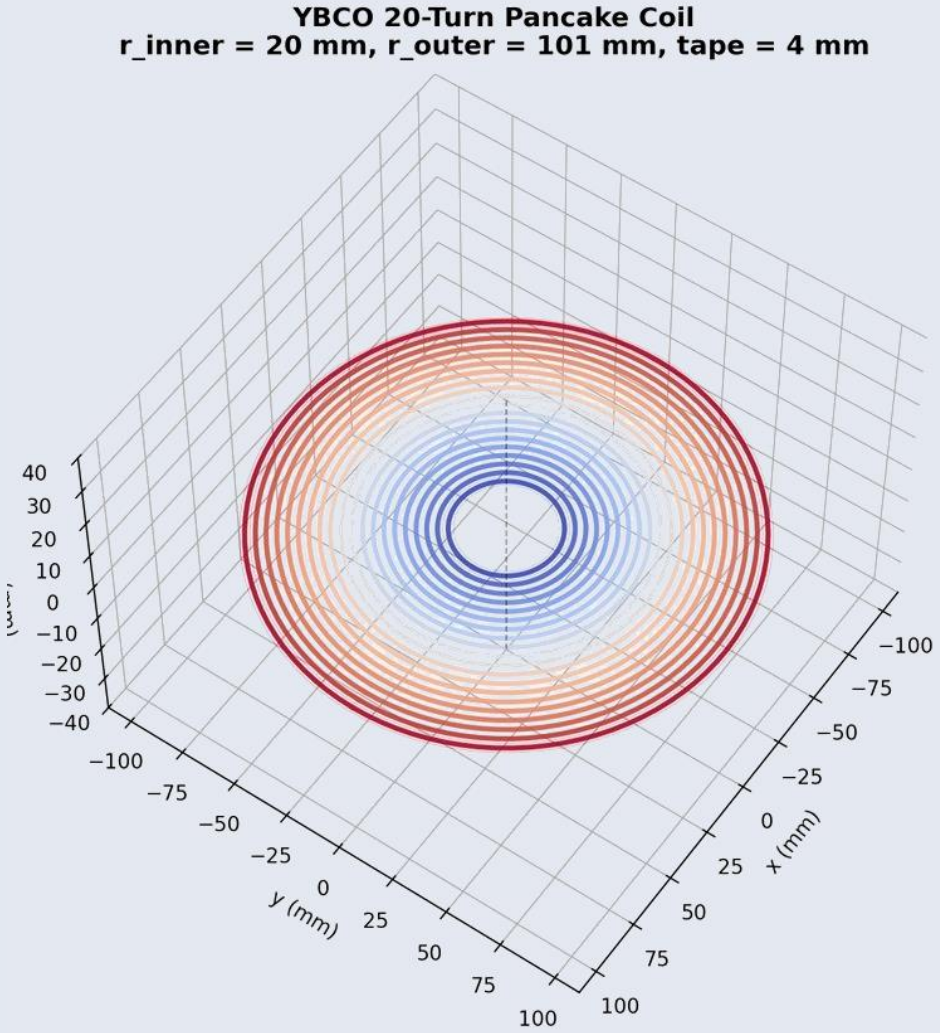


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Electro-Magnetic Superconducting De-Tumbler

- YBCO high-temperature superconductor pancake coil
- Contactless eddy-current braking: **no risk of fragmentation during detumbling**
  - One of our key mission requirements
- Magnetic dipole projected at target induces braking torque in conductive structure of satellite
- Operating envelope: 800 A safe at 77 K
- Dipole moment of 4,200 A*m^2 produces ~0.15 mT at 2 m standoff
- Cryocooler required for high-current (> 2kA) operation

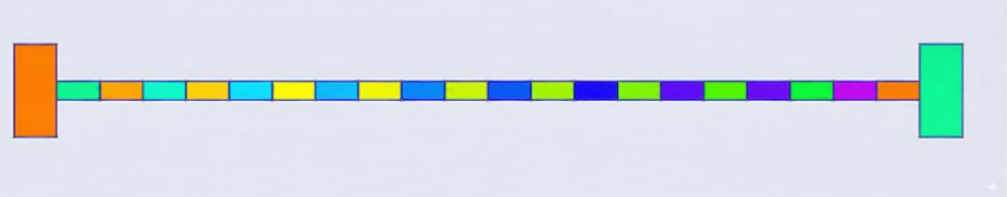


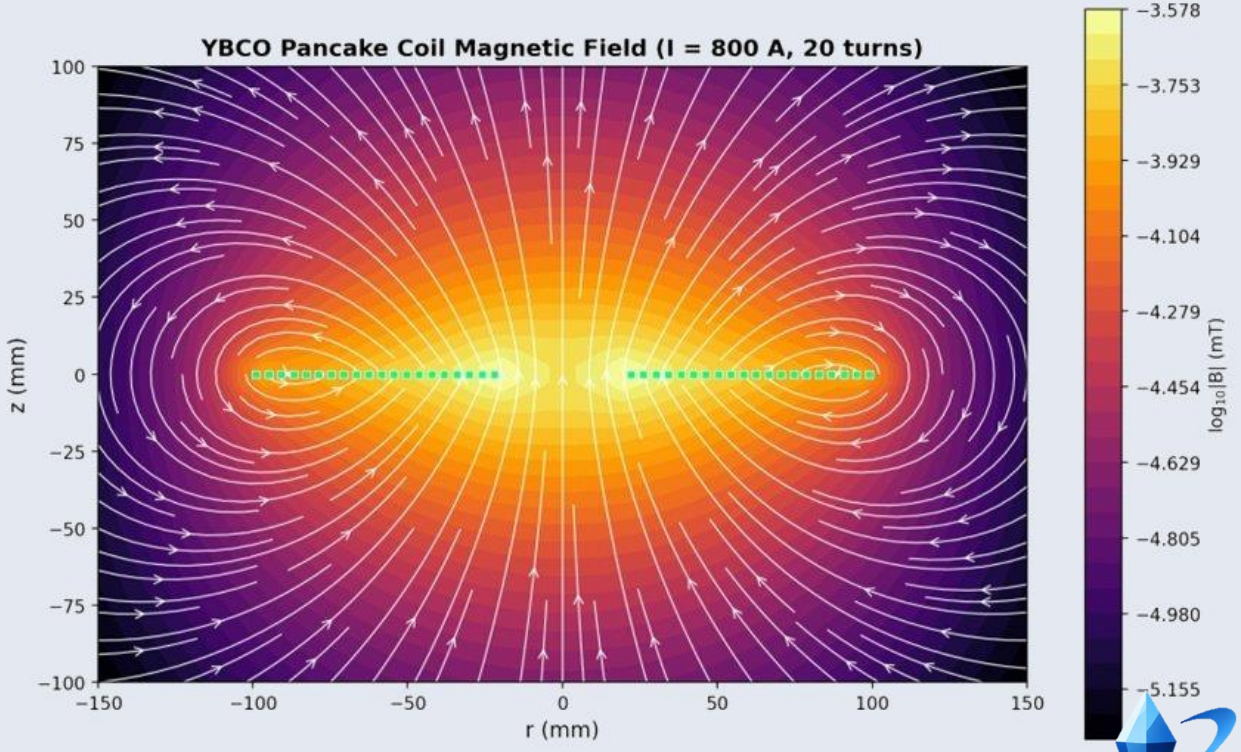


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Coil Design

- 20-turn YBCO pancake coil 10 cm Diameter
- Tape stack: YBCO superconductor / Hastelloy substrate / Cu stabilizer
- T_c = 92 K, superconducts well above 4 K space background
- Field simulation confirms dipole-like far-field behavior
- Field concentrated in coil plane, decays with $\frac{1}{r^3}$ at axial distance z
- At 800 A: strong enough to induce eddy currents at 1.5-3 m

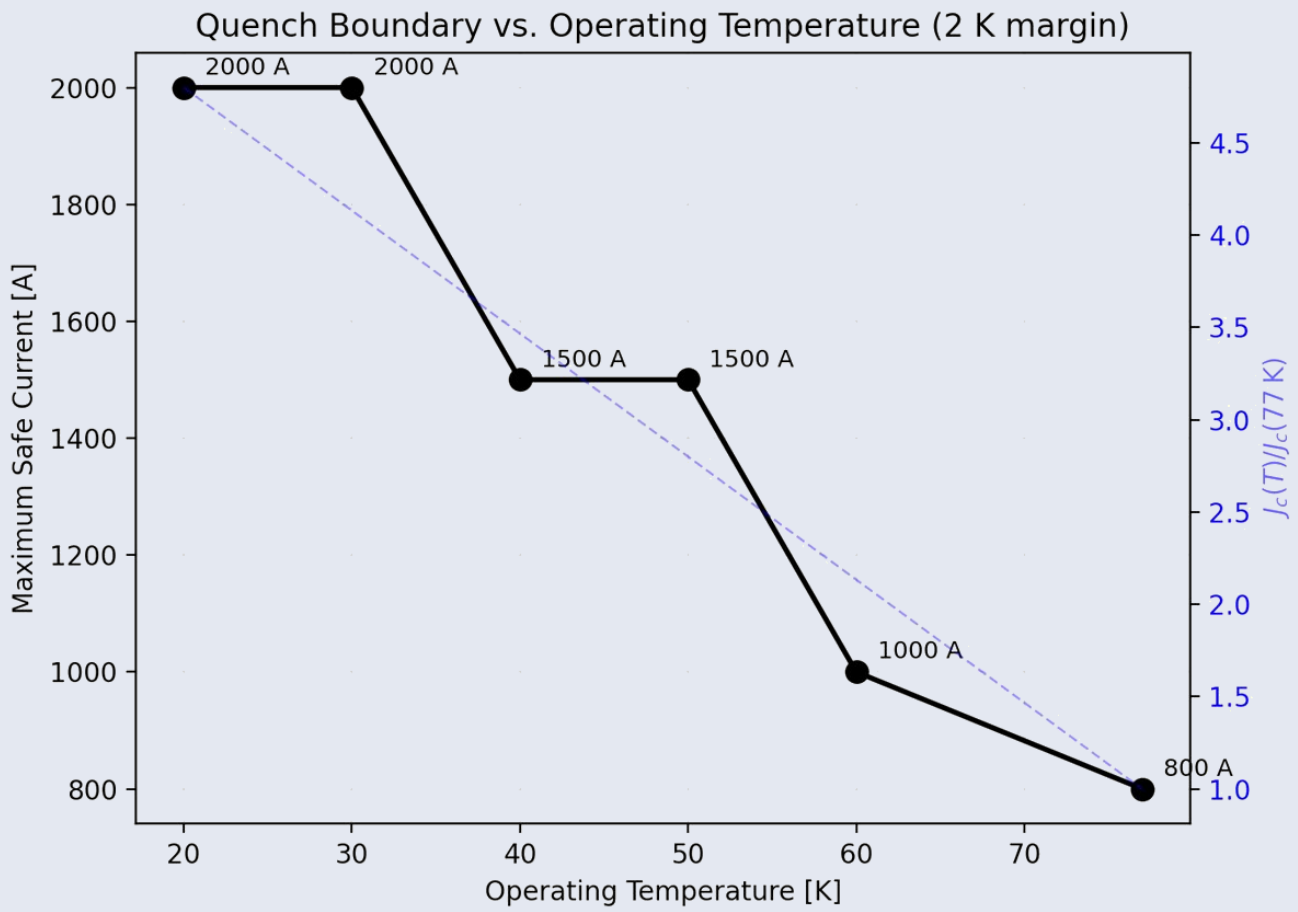




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Quench Determination

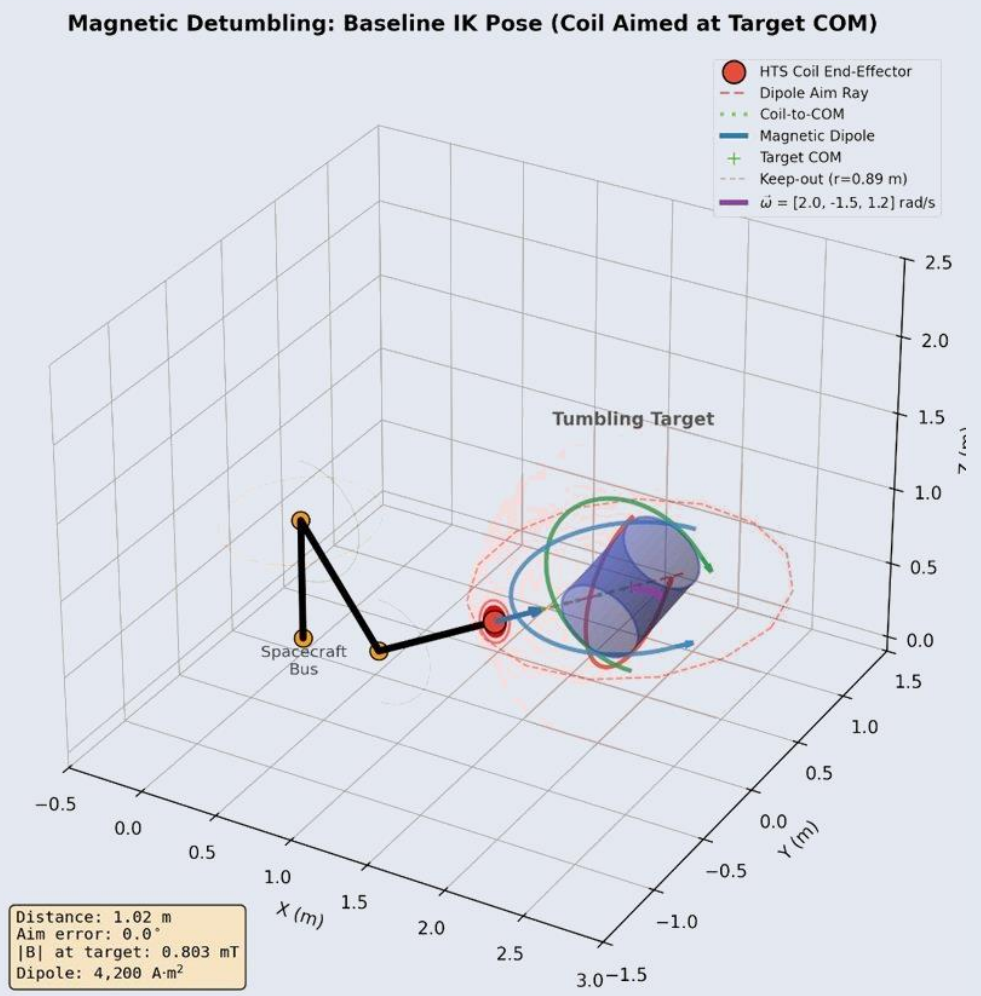
- Analytical model and in progress FEA model allow for quench simulation
- Unlikely that a cryocooler is needed since we can produce large B-field at radiative cooling temperatures



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Detumbling Mechanics

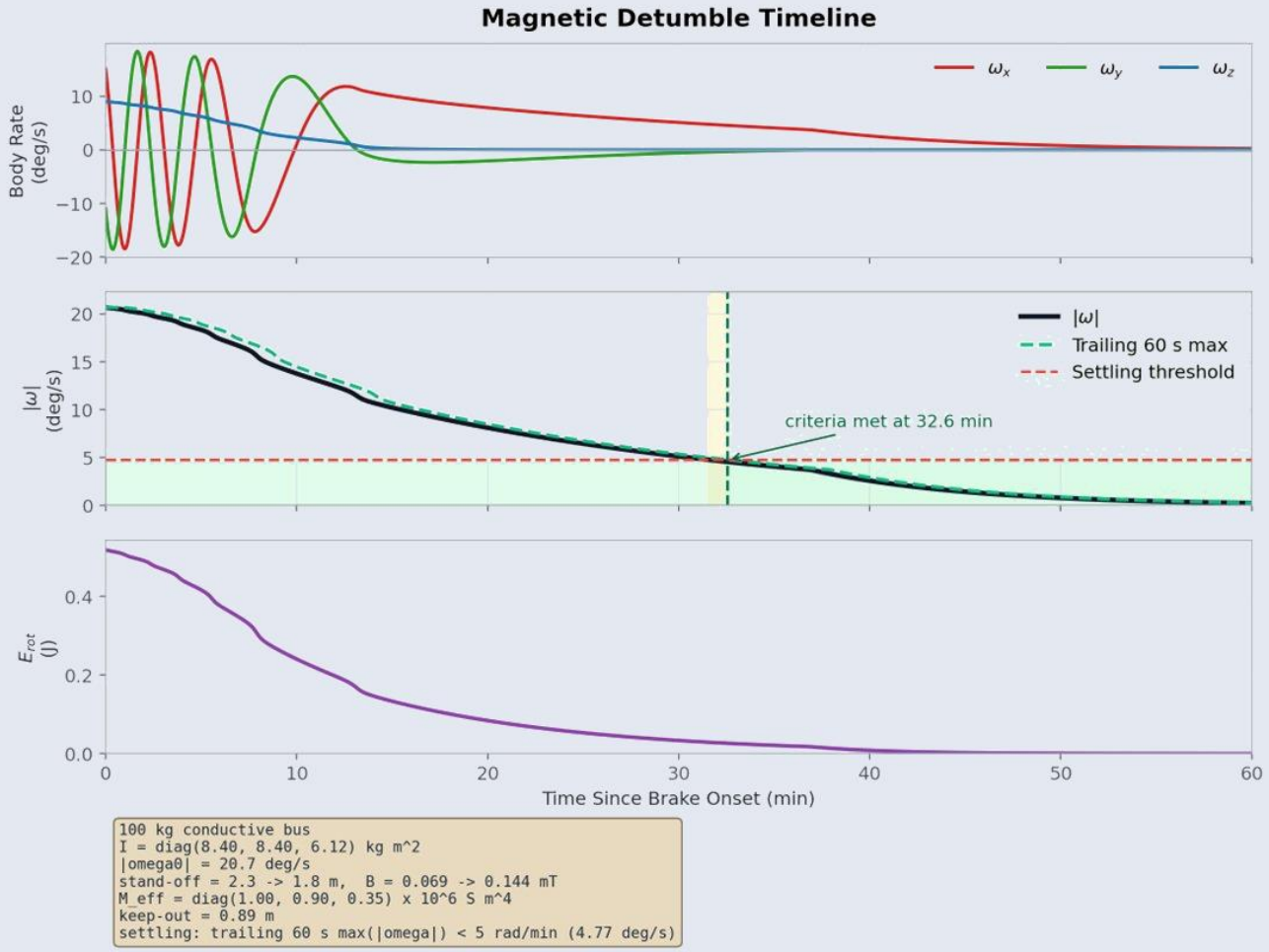
- A 3 DOF robot arm allows optimal pointing for minimum detumbling time
- Generally pointing at the COM gives the fastest time to detumble
- A reinforcement learning algorithm (DDPG) was trained to point the electromagnet at the most optimal position over time



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Detumbling Time Estimate

- Detumbling Criteria is less than 5 deg/s
- A 100kg debris with a reasonable Magnetic Tensor and Moment of Inertia meets criteria in **32.6 minutes**
- Larger/heavier objects take days

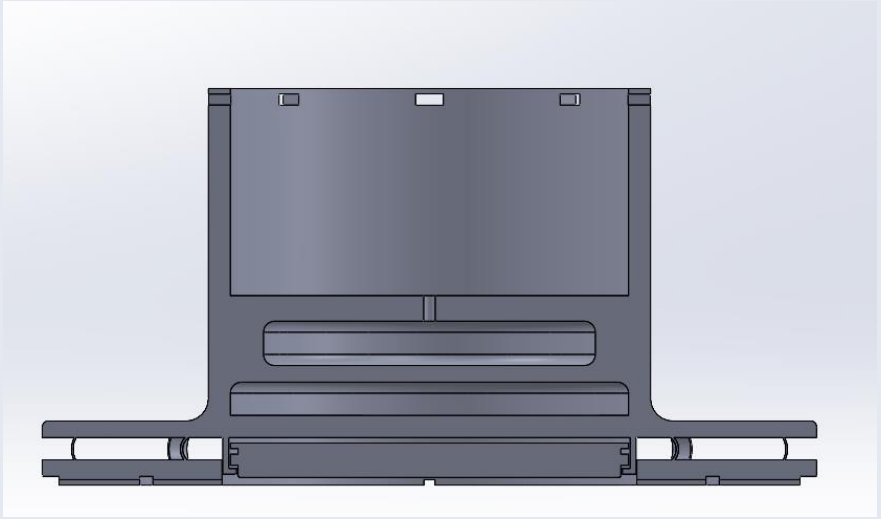


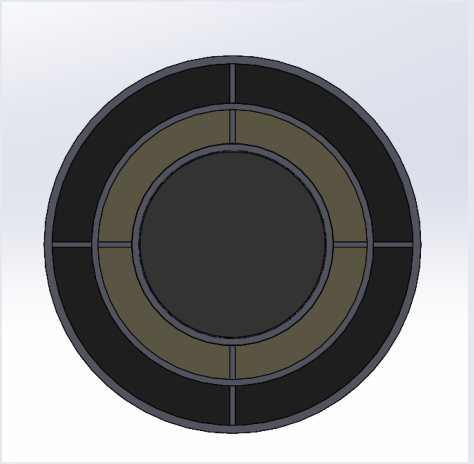
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GOLD for De-Orbiting

Gossamer Orbit Lowering Device (GOLD)

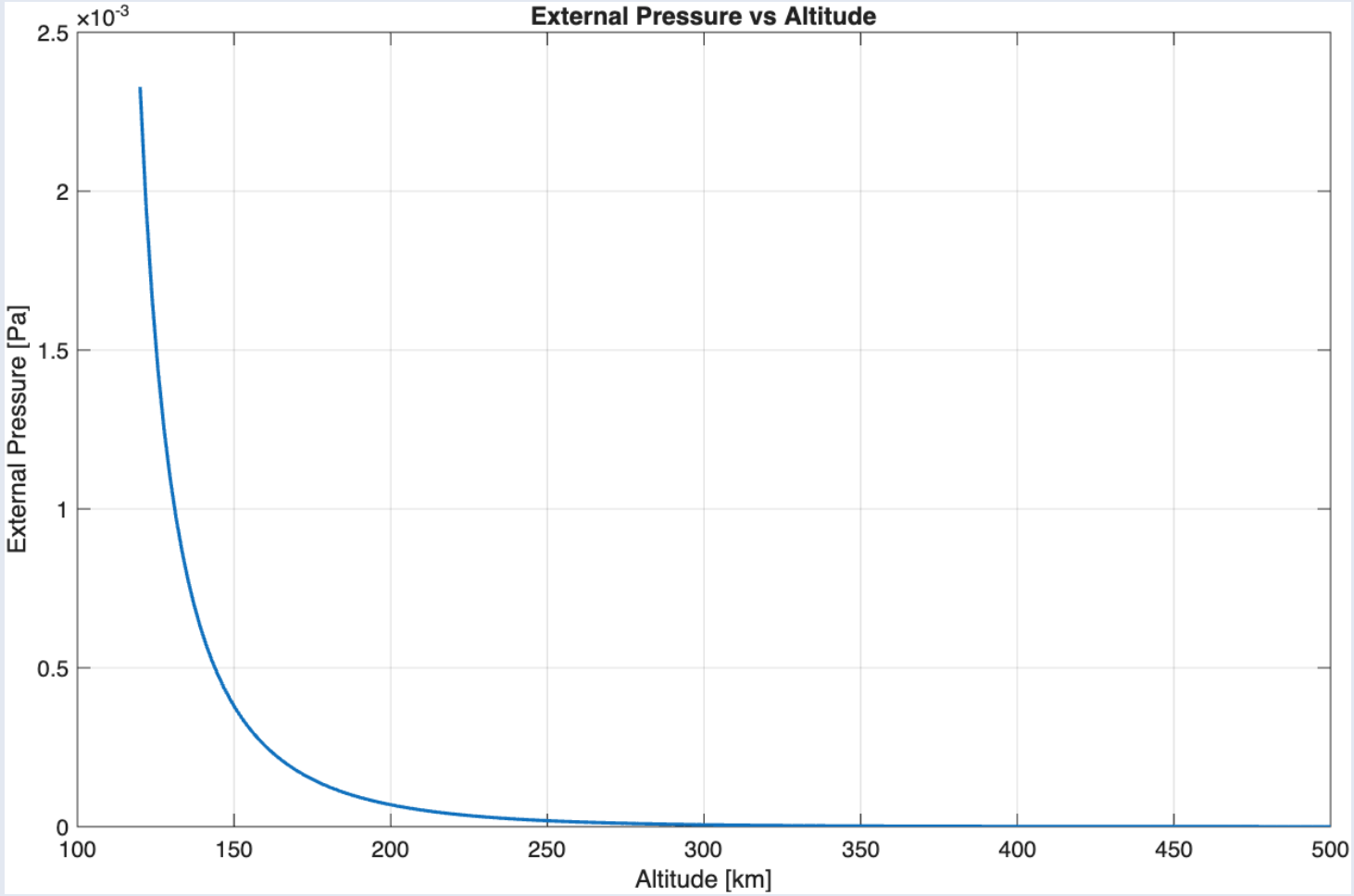
- An attachment package will extended from satellite carrying GOLD on a boom arm
- A temporary Vander Waal adhesive will adhere the GOLD system with low strength while thermoplastic adhesive activates
- A thermoplastic adhesive that will adhere once softened and conformed to the spacecraft surface when heated to 20 °C
- Once the thermoplastic has set, a Jetweld adhesive pad will be heated to 149 °C and applied to the surface with 14 kPa pressure for permanent adhesion with a strength of 34 kPa





GOLD for De-Orbiting

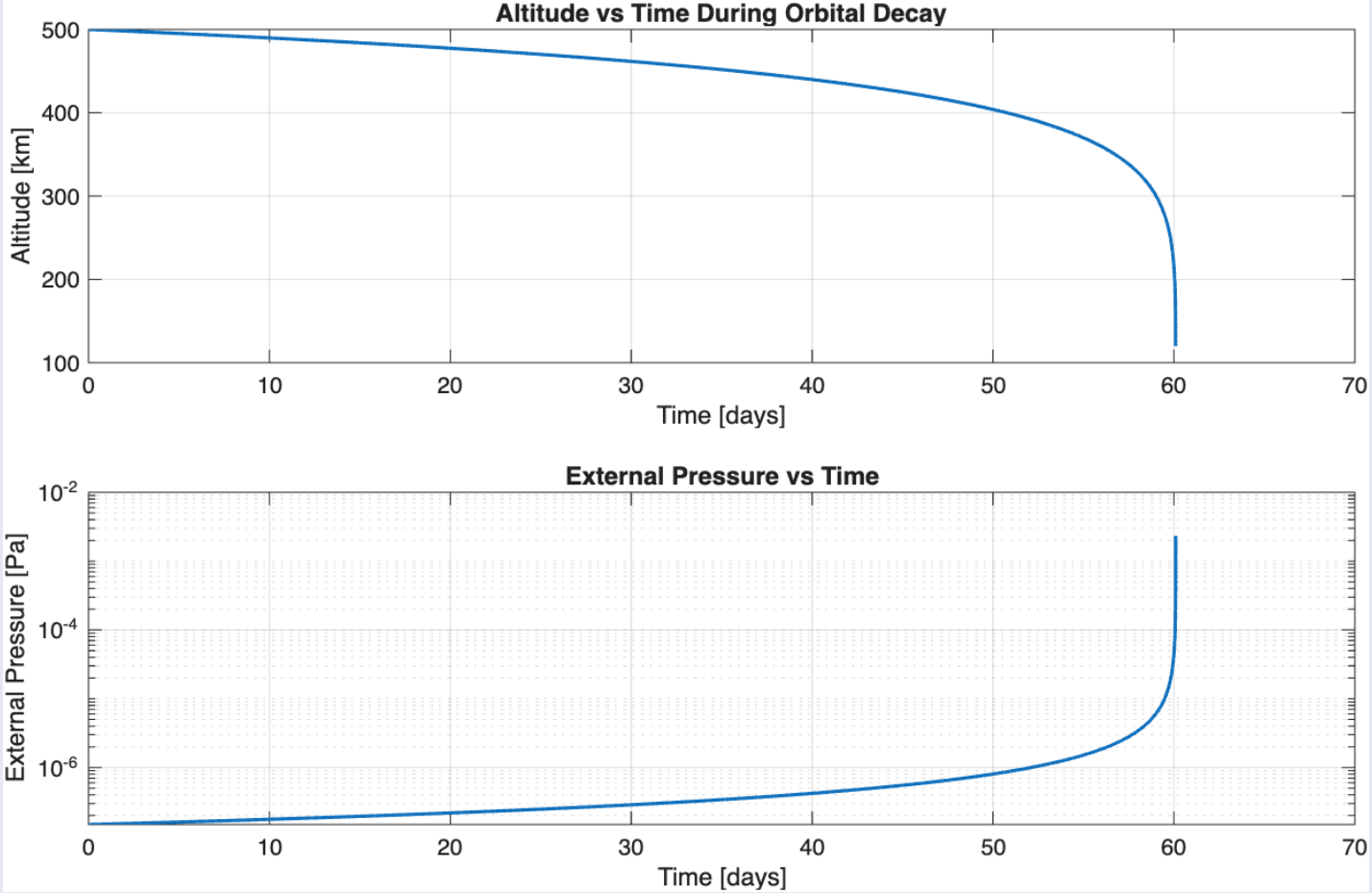
- An onboard micro controller triggers an actuator to break the seal between the vacuum balloon and solid sublimating material
- The material slowly sublimates, producing gas to inflate the balloon to 95.5 m^3
- At 0.05 Pa internal pressure, the balloon expands to a 2.84 m radius, increasing drag area
- The 3.6 kg balloon uses thin Kapton film for lightweight strength and thermal stability
- A diamond-grid ripstop reinforcement helps the membrane withstand pressure and prevent tear growth



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G.O.L.D. for De-Orbiting

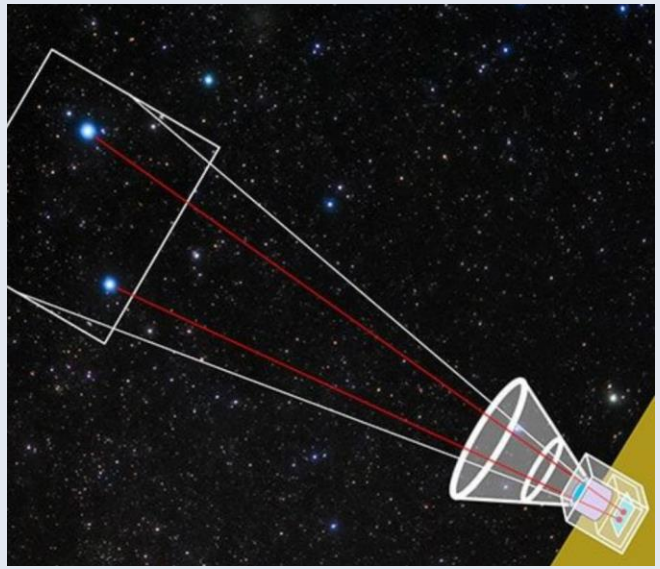
- Example:
- A 100 kg target satellite
- Deorbit in 60 days from 500 km to 120 km where it will deorbit on its own
- Maintain internal balloon pressure higher than external pressure for shape



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ACDS

- Attitude Control
  - Angular momentum transfer from the target can disturb spacecraft attitude
  - **Reaction wheels** provide stabilization
  - **Monopropellant thrusters** provide large moments as needed for large maneuvers and momentum dumping
  - **Magnetic torquers** are being studied for use in between targets to dump momentum
- Attitude and Position Determination
  - **Star tracker and GPS** provide absolute information.
  - **IMU** propagates between readings



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GNC

Components Budgeting



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Components Sizing

Component	Features	Power Requirements	Weight	Cost
Small Sat PMAD (Endurosat)	- Integrated Battery subsystem - 7000W Max Power - No Quote Needed - TRL9	- NA		- $310000
BHT 600 Hall Effect Thruster System (Busek)	- 150 mn Max Thrust - TRL9 - Full Integrated System	- 8000W	- 15kg (dry) - 55kg (wet)	- ~$2M
Hydrazine MP System (Moog)	- Full Integrated System	- 200W when firing		- ~$1.5M
RW4 Reaction Wheel (Blue Canyon)	- 8 nm Max Momentum - Max Torque .25 nm - 4 total	- 176W (PP)	- 4.4kg	- ~$80,000 PP

Components Sizing

Component	Features	Cost	Weight	Power Requirements
Solar Panels	- 3 Panels Per Wing - 3.5A Per Section - Maximum Open Circuit Voltage: 92.2V	- $500,000	Depends on Required Area, Then Using Mass formula: M = 3(3.41kg/m^2 * AP(Panel Area) + 0.65 kg) + 0.4kg	N/A
Three and Six Degree of Freedom Boom Arms	- Will need to be developed in house	- 6 DOF: $2.5 M - 3 DOF: $1.25 M	N/A	N/A
Primary Structure	- Over 12U Payload Envelope - Customizable Payload Mass and Average Orbital Power Usage	- $600,000	Depends on Chosen Specifications	Depends on Chosen Specifications
Thermal Control System	- Environment: -40 C to +70 C - ~ 87,000 Hour Lifespan	- $ 2.5 M	4.5 kg (Single Stage Cooler) 3.8 kg (Electronics)	180 W

Conclusion

- SCRAP aims to improve the condition of LEO by removing pieces of debris from orbit.
  - Through systematic analysis, ideal candidates are chosen.
  - De-tumbling is conducted to stabilize the target.
  - De-orbiting begins with the attachment of the GOLD System.
  - The GOLD System then increases drag and decays the orbit.
- Due to the massive increase in LEO congestion, regulatory bodies and private entities are going to have to move to more sustainable space practices.
  - A solution to this problem will *need* to become available, and we are here.

Risk Assessment & Feasibility

Superconductor Quenching (1)

- Operating temperature must remain below 92K at all times
- Always maintain current density below critical current density (J_c)
- Always maintain applied field below critical magnetic field (H_c)
- J_c and H_c limits sourced from manufacturer specifications

Mitigation Strategies

- Solar radiation insulation added around SuperCon Coils
- Use of Liquid Helium Cryo-Cooler (tentative)

Magnetic Field interacts with bus components (2)

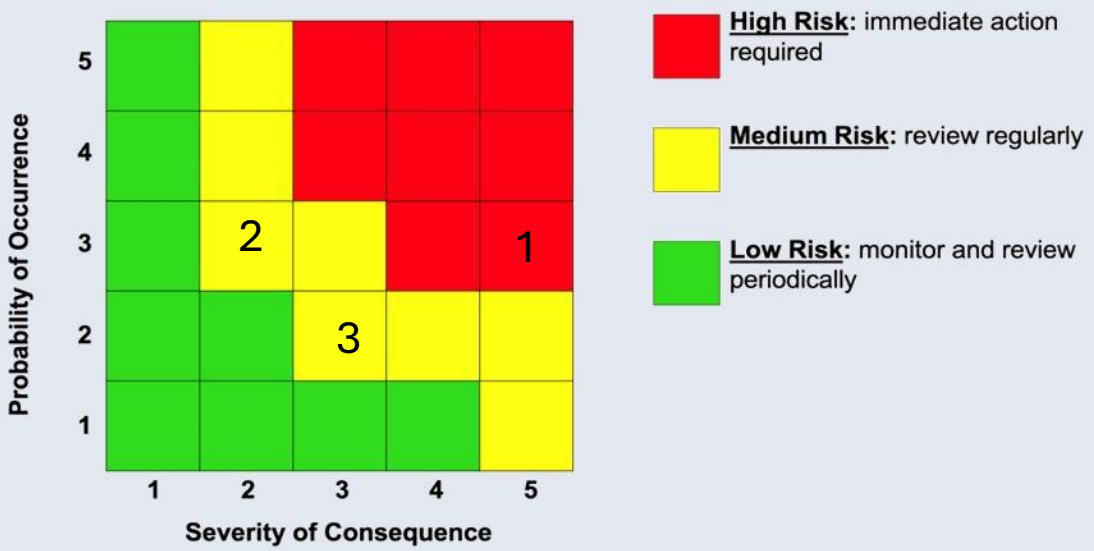
Mitigation Strategies

- Coat important materials in MuMetal
- Space the electromagnet from the bus body with a boom

Gold Solution becomes detached during de-orbit (3)

Mitigation Strategies

- Extensive vetting of ideal attachment position for the GOLD system
    - Searching out large flat areas on the target satellite



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